QUOC BUI ANH

IC LAYOUT ENGINEER

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SUMMARY

Final-year Electrical and Electronic Engineering student with hands-on experience in IC layout, layout verification through academic projects. Proficient in Cadence Virtuoso and familiar with the IC/ASIC design flow, including DRC/LVS verification and physical layout considerations. Detail-oriented and motivated to further develop my expertise in full-custom IC layout and semiconductor design.

EDUCATION

Ho Chi Minh City University of Technology (HCMUT) Sep 2022 – Apr 2027
Bachelor of Engineering in Electrical and Electronic Engineering
Major: Electronics and Telecommunications Engineering
Availability: Full-time

TECHNICAL SKILLS

IC Design & Layout: IC Layout, Digital Circuit Design, Layout Verification, DRC/LVS Debugging, Post-Layout Simulation
Layout Techniques: Device Matching, Symmetry, Common-Centroid, Interdigitation, Dummy Devices
Layout & Reliability Knowledge: LOD, WPE, Antenna Effect, Latch-up, Electromigration
EDA Tools: Cadence Virtuoso, Cadence PVS, Cadence Spectre, Cadence Xcelium, Cadence Genus
RTL & FPGA: SystemVerilog, Intel Quartus, ModelSim/QuestaSim, FPGA Implementation
Programming & Platforms: C, Linux

WORK EXPERIENCE

FPGA Design Intern | Gowin Semiconductor & OneKiwi Technology June 2025 – Sep 2025

- Developed and verified FPGA designs using GOWIN Designer and QuestaSim on the ACG525 development board, including functional simulation and on-board validation.
- Designed and implemented RTL modules including LED counters, UART transmit/receive modules for single- and multi-byte communication, and FSM-based sequence detectors.
- Designed and implemented a Modbus RTU communication IP core for data exchange between a Linux host and peripheral devices, enabling monitoring and control of digital I/O in a wastewater treatment system; deployed and validated on a GW1N-4K FPGA.

PROJECT 🔗

Digital IC Design & Layout – 45 nm PDK

Cadence Virtuoso, Cadence PVS, Cadence Xcelium, Cadence Genus

- Designed transistor-level schematics and custom physical layouts for digital building blocks, including logic gates, multiplexers, a multi-stage tapered buffer, and a custom transistor-level D flip-flop.
- Designed and implemented 8×8 SRAM and 8×8 TCAM circuits, including schematic design, functional verification
- Performed DRC and LVS verification using Siemens Calibre and debugged layout-rule violations and schematic-to-layout mismatches.
- Conducted functional simulations to verify circuit behavior before physical implementation and validated layout correctness through DRC/LVS.

All-Digital Delay-Locked Loop (ADDLL) – In Progress

Cadence Virtuoso, Cadence Spectre

- Designed and simulated an all-digital delay-locked loop architecture incorporating a digitally controlled delay line (DCDL) and fast-lock control scheme.
- Verified schematic-level operation up to 7 GHz and evaluated circuit functionality and timing across multiple process, voltage, and temperature (PVT) corners.
- Conducted functional and post-layout verification to validate circuit operation after physical implementation.

LFSR Project - LFSR-Based Random Number Generator Feb 2025 – June 2025

SystemVerilog, Quartus, Cadence Virtuoso, Genus, Xcelium

- Designed an LFSR-based pseudo-random number generator incorporating polynomial selection, counters, multiplexers, and comparison logic.
- Implemented and verified the digital architecture through simulation and synthesis.
- Evaluated generated random sequences using statistical tests based on the NIST SP 800-22 test suite.

LANGUAGES

Vietnamese - Native

English - TOEIC L&R: 690 🔗 (2026)

WORK EXPERIENCE

Hardware Assembler (Part-time) | ITR VN July 2025 – Oct 2025 (Project-based)

- Assembled and tested electronic hardware used for heart-rate and respiration measurement systems.
- Supported PCB troubleshooting, component replacement, and functional testing of prototype hardware.